Learning Speech Representations via Self-Supervised Learning — wav2vec 2.0

Baevski et al. (2020), in their paper *wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations*, present a framework for Self-Supervised Learning of speech representations from raw audio signals, without the need for transcription at the pre-training stage. The study addresses the dependence of Automatic Speech Recognition (ASR) systems on large amounts of labeled data, and proposes exploiting unlabeled audio datasets to learn representations that can subsequently be adapted to supervised tasks. This approach makes it possible to separate the learning of speech features from the learning of the target task, and may thereby reduce the amount of labeled data required.

The architecture combines a Convolutional Feature Encoder that extracts continuous latent representations from the audio signal, and a Transformer that produces contextual representations by means of learning relations between speech segments. During the Pre-training, segments of the latent representations are masked, and the model is trained by means of Contrastive Learning to identify the correct target representation of the masked segment out of a set of candidates. The training targets are produced by means of a Quantization Module based on Product Quantization and Gumbel-Softmax, and are learned jointly with the other components of the model. In addition, a Diversity Loss is used to encourage diverse use of the discrete codes. After the pre-training, the researchers carry out Fine-Tuning on transcribed speech data by means of a Connectionist Temporal Classification (CTC) Loss for speech recognition tasks.

The study was evaluated using the LibriSpeech and LibriVox datasets, examining BASE and LARGE model configurations and varying amounts of labeled training data. In an experiment in which a LARGE model underwent Pre-training on about 53,200 hours of unlabeled audio and was then Fine-Tuned using only ten minutes of labeled data, a Word Error Rate (WER) of 4.8% was achieved on test-clean and 8.2% on test-other, using a language model at decoding time. When 960 hours of labeled data were used, results of 1.8% and 3.3% were obtained, respectively. In addition, a comparison with training from scratch demonstrated an improvement in performance following Pre-training even under conditions in which a large amount of labeled data is available. Ablation experiments showed that using continuous representations as input to the Transformer and Quantized targets for the Contrastive Loss achieved the best results among the alternatives that were examined, thereby supporting the central architectural choice of the method.

Despite the study's achievements, it is important to emphasize that wav2vec 2.0 was originally developed and evaluated for Speech Recognition and Phoneme Recognition, and not for voice forgery detection. Accordingly, the results do not prove that the learned representations contain all the acoustic features required for detecting synthetic speech or voice manipulations. Moreover, representations adapted for recognizing linguistic content may be sensitive to features different from those used to distinguish between real and fake speech. In addition, the original Pre-training requires extensive computational resources, and some of the speech recognition results also rely on external language models.

For the present study, the central contribution of wav2vec 2.0 is in presenting an infrastructure for learning speech representations that can be adapted to new tasks even when the amount of labeled data is limited. Given the relatively small recording dataset in the voice forgery detection project, it is possible to examine using a pre-trained model as a frozen Feature Extractor, and to compare it with a Fine-Tuning approach in which part of its weights are adapted to the classification task. In addition, its performance can be compared with models based on MFCC, CQCC and AASIST, using a uniform evaluation protocol and EER as the central metric. This comparison will make it possible to examine whether speech representations learned by means of Self-Supervised Learning provide information useful for forgery detection and for the ability to generalize between different data sources and attack methods.

Bibliographic source:

Baevski, A., Zhou, H., Mohamed, A., & Auli, M. (2020). *wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations*. arXiv:2006.11477. https://arxiv.org/abs/2006.11477